Derangement

Derangement is an arrangement of objects in which no object is in its original position. For Example, a permutation of the integers $1, 2, 3, \cdots, n$ in which 1 is not in the first place, 2 is not in the second place, 3 is not in the 3rd place, and so on and n is not in the nth place is a derangement.

The no. of possible derangements of 'n' distinct objects 1, 2, 3, --, n is denoted by d_n. If there is only one object, it continues to be in its original place in every arrangement; then $d_1 = 0$. If there are two objects, a derangement can be done in only one way - by interchanging their places; then $d_2 = 1$. For three objects 1, 2, 3, the possible derangements 231 and 312, then $d_3 = 2$.

In general, the no. of derangements of 'n' distinct objects is

$$d_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} \quad \text{— (1)}$$

Note: For $n \geq 7$, the formula in eq (1) can be rewritten as

$$d_n = \frac{n!}{e} = n! \times e^{-1} = [n! \times 0.3679] \quad 0.3679$$

$d_1 = 0$; $d_2 = 1$; $d_3 = 2$, $d_4 = 9$; $d_5 = 44$, $d_6 = 265$,
$d_7 = 1854$, $d_8 = 14833$ - - - -

(i) Find the number of derangements of 1, 2, 3, 4.

Let S be the set of all arrangements of 1, 2, 3, 4

$\therefore |S| = N = S_0 = 4!$

Let c_1, c_2, c_3 and c_4 denote the 1 is in the right place, 2 in the right place, 3 in right place and 4 in the right place respectively.

By the P. of I and E, we have

$$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4$$

we have $S_1 = \Sigma$

$N(C_1) = 3! \Rightarrow S_1 = {}^4C_1 \times 3!$

$S_2 = \Sigma N(C_i C_j)$

$N(C_1 C_2) = 2! \Rightarrow S_2 = {}^4C_2 \times 2!$

$S_3 = \Sigma N(C_i C_j C_k)$

$N(C_1 C_2 C_3) = 1! \Rightarrow S_3 = {}^4C_3 \times 1!$

$S_4 = \Sigma N(C_i C_j C_k C_l)$

$N(C_1 C_2 C_3 C_4) = 0! \Rightarrow S_4 = {}^4C_4 \times 0! = 1$

$\bar{N} = 4! - ({}^4C_1 \times 3!) + {}^4C_2 \times 2! - ({}^4C_3 \times 1!) + 1 = 9$

(2) In how many ways 1, 2, 3, ---, 10 be arranged on a line so that no even integer is in its right place.

Let S be the set of all arrangements of the given digits from 1, 2, 3, ---, 10.

$\therefore |S| = N = S_0 = 10!$

Let C_i, where $1 \le i \le 5$, denotes 2, 4, 6, 8, and 10 in the right place respectively.

By the P. of. i and e, we have

$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4 - S_5$

$S_1 = \Sigma N(C_i)$

$N(C_1) = 9! \Rightarrow S_1 = {}^5C_1 \times 9!$

$S_2 = \Sigma N(C_i C_j) \Rightarrow N(C_1 C_2) = 8!$

$S_2 = {}^5C_2 \times 8!$

Similarly $S_3 = {}^5C_3 \times 7!$ $S_4 = {}^5C_4 \times 6!$ and $S_5 = {}^5C_5 \times 5!$

$\bar{N} = 10! - ({}^5C_1 \times 9!) + ({}^5C_2 \times 8!) - ({}^5C_3 \times 7!) + ({}^5C_4 \times 6!) - ({}^5C_5 \times 5!)$

Note :- $N - \bar{N}$ gives the no. of arrangements ... one of the conditions

⑦ Let $A = \{1, 2, \ldots, 7\}$, $f: A \to A$ is said to have fixed point if for some x in A, $f(x) = x$. How many one-to-one functions $f: A \to A$ have at least one fixed point.

▲ These are $7!$ one-to-one functions are there from $A \to A$.

Here, the no. of derangements $d_7 = \bar{N}$

∴ No. of functions with at least one fixed point

$= 7! - d_7$.

⑧ A Machine that inserts letters into envelopes goes haywire and inserts letters randomly into envelopes. what is the probability that in a group of 100 letters @ no letter is put into the correct envelope ⓑ exactly 1 letter is put into the correct envelope ⓒ exactly 98 letters are put into the correct envelope ⓓ exactly 99 letters are put into correct envelope ⓔ all letters are put into the correct envelopes?

▲ Note :- Before problem

The probability of derangement of n objects is $\frac{d_n}{n!}$.

For example:

n:	2	3	4	5	6	7
$\frac{d_n}{n!}$:	0.5	0.333	0.375	0.3667	0.36806	0.36786

ⓐ The probability of no letter being put in the correct envelope is $\frac{d_{100}}{100!}$. Because the no. of favorable cases (i.e derangements) is d_n and the total no. of favorable cases is $n! = 100!$

ⓑ When exactly 1 letter is put correctly, the no of derangements for the remaining 99 letters is D_{99}. This can happen in $100c_1$ ways. So the probability is $\frac{100c_1 \times D_{99}}{100!}$

Rook Polynomials :-

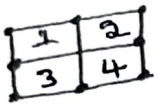
A rook polynomial is a polynomial

$$r(C,x) = 1 + r_1 x + r_2 x^2 + \cdots + r_n x^n,$$

where C is the chess board and r_n denote the no. of ways in which 'n' rooks (i.e pawns) can be placed on a board such that no two rooks are in the same row or in the same column of the board.

Find the rook polynomials for the following chess boards:

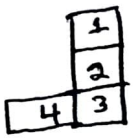
(i)



Here $r_0 = 1$, $r_1 = 4$, $r_2 = 2$

$\therefore r(c,x) = 1 + 4x + 2x^2$

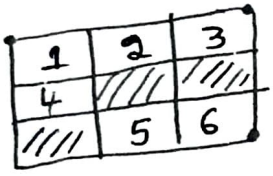
(ii)



Here $r_0 = 1$, $r_1 = 4$, $r_2 = 2$

$r(c,x) = 1 + 4x + 2x^2$

(iii)

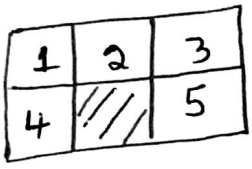


Here $r_0 = 1$, $r_1 = 6$, $r_2 = 8$, $r_3 = 2$

(1,5), (1,6), (2,4), (3,4) (3,4,5) (2,4,6)
(2,6), (2,6) (4,5), (4,6)

$r(c,x) = 1 + 6x + 8x^2 + 2x^3$

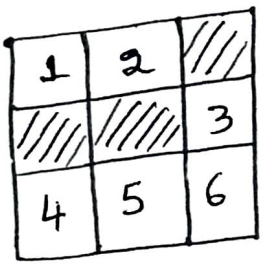
(IV)



$r_0 = 1$, $r_1 = 5$, $r_2 = 4$

$r(c,x) = 1 + 5x + 4x^2$

(V)

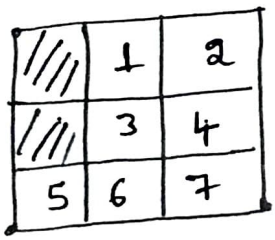


(1,3), (1,5), (1,6)
(2,3) (2,4), (2,6)
(3,4), (3,5)
(1,3,5) (2,3,4)

Here $r_0 = 1$, $r_1 = 6$,
$r_2 = 8$, $r_3 = 2$

$r(c,x) = 1 + 6x + 8x^2 + 2x^3$

(VI)

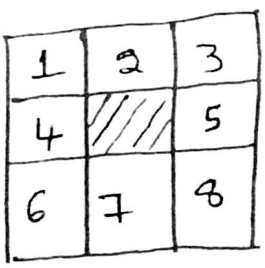


(1,4), (1,5), (1,7)
(2,3), (2,5), (2,6)
(3,5), (3,7), (4,5)
(4,6)
(2,3,5) (1,4,5)

Here $r_0 = 1$, $r_1 = 7$, $r_2 = 10$
$r_3 = 2$

$r(c,x) = 1 + 7x + 10x^2 + 2x^3$

(VII)



(1,5) (1,7) (1,8)
(2,4) (2,5) (2,6) (2,8)
(3,4) (3,6) (3,7)
(4,7) (4,8) (5,6) (5,7)
(1,5,7) (2,4,8) (3,4,7)
(2,5,6)

Here $r_0 = 1$, $r_1 = 8$,
$r_2 = 14$, $r_3 = 4$

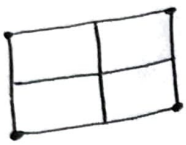
$r(c,x) = 1 + 8x + 14x^2 + 4x^3$

Expansion Formula :- On a given board C, suppose we choose a particular square and mark it ⊛. Let D be the board obtained from C by deleting the row and column containing the square ⊛, and let E be the board obtained from C by deleting only the square ⊛. Then, the rook polynomial $r(C,x)$ for the board C has the following property:

Property: $r(C,x) = x\, r(D,x) + r(E,x)$

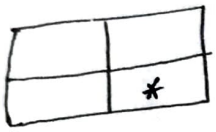
This is known as the expansion formula for $r(C,x)$.

Ex:



$r(C,x) = 1 + 4x + 2x^2$

C:

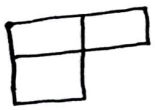


$\Rightarrow$ D:



$r(D,x) = 1 + x$

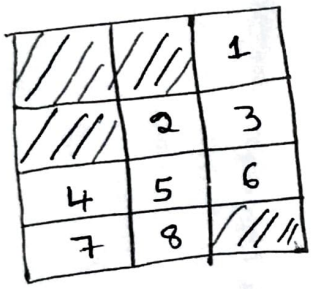
E:



$r(E,x) = 1 + 3x + x^2$

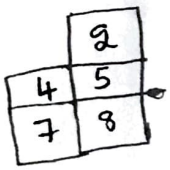
$\therefore r(C,x) = x\, r(D,x) + r(E,x) = 1 + 4x + 2x^2$.

① Using the Expansion formula, obtain the rook polynomial for the board C shown below:

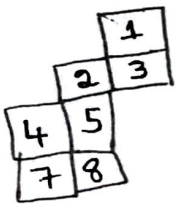


▶ Let us mark the square 6 as in the given board as ⊛.

D:

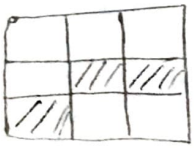


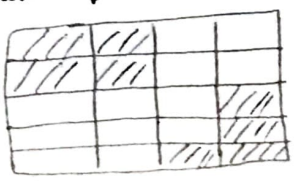
E:



Ans $r(C,x) = 1 + 8x + 16x^2 + 7x^3$

② Find the rook polynomial for the following:





Suppose a board C is made up of two parts C_1 and C_2 where C_1 and C_2 have no squares in the same row or column of C, such parts of C are called disjoint sub-boards of 'C'. Then the rook polynomial $r(C,x)$ for the board C has the following property: $r(C,x) = r(C_1,x) \times r(C_2,x)$.

3×3 → Rook polynomial. do

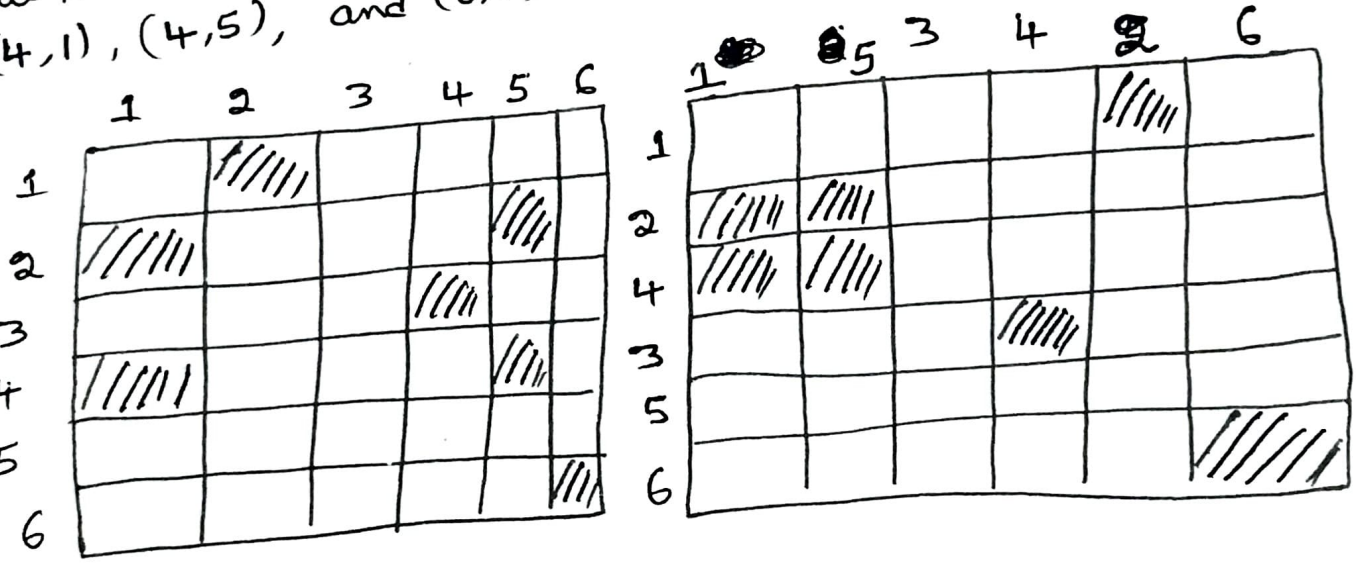
Derangements :-

① . Ms. Pezzulo teaches geometry and then biology to a class of 12 advanced students in a classroom that has only 12 desks. In how many ways can she assign to these desks so that (a) no student is seated at the same desk for both classes ? (b) these are exactly six students each of whom occupies the same desk for both classes ?

(a) $12! \times d_{12}$ (b) $12! \times {}_{12}C_6 \times d_6$.

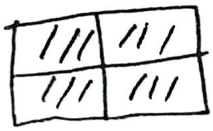
Rook Polynomials

① - We have a pair of dice ; one is red, the other green. We roll these dice six times. What is the probability that we obtain all six values on both the red die and the green die, if we know that the ordered pairs (1,2), (2,1), (2,5), (3,4), (4,1), (4,5), and (6,6) did not occur ?



C_1 : 

$r(C_1, x) = 1 + x$

C_2 : 

$r(C_2, x) = 1 + 4x + 2x^2$

C_3 : 

$r(C_3, x) = 1 + x$

C_4 : 

$r(C_4, x) = 1 + x$

$$r(C, x) = r(C_1, x) \times r(C_2, x) \times r(C_3, x) \times r(C_4, x)$$
$$= (1+x)(1+4x+2x^2)(1+x)(1+x)$$
$$= (1+4x+2x^2)(1+x)^3$$
$$= 1 + 7x + 17x^2 + 19x^3 + 10x^4 + 2x^5.$$

Here $n=6$, $r_0=1$, $r_1=7$, $r_2=17$, $r_3=19$, $r_4=10$, $r_5=2$, $r_6=0$.

$s_0=6!$ $s_k=(n-k)!\times r_k=(6-k)!\times r_k$

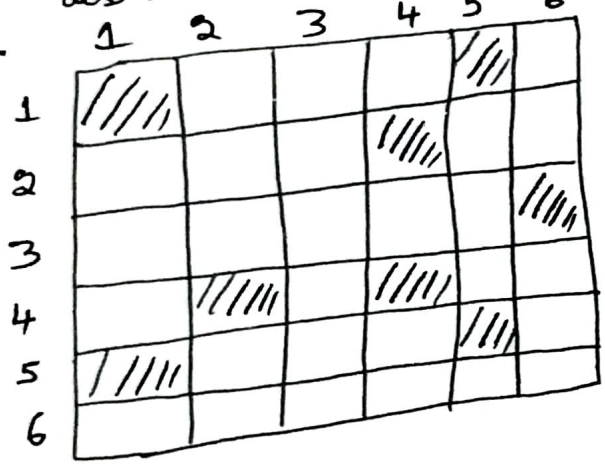
$s_1=5!\times 7$; $s_2=4!\times 17$; $s_3=3!\times 19$; $s_4=2!\times 10$;

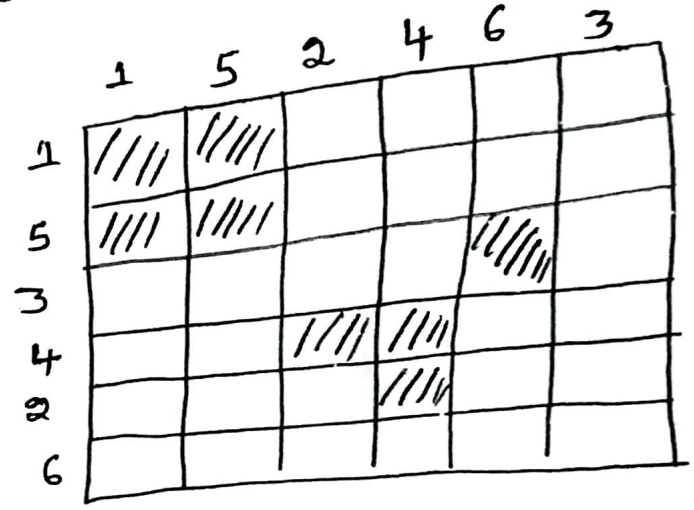
$s_5=1!\times 2$; $s_6=0!\times 0=0$.

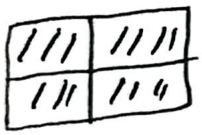
$\bar{N}=s_0-s_1+s_2-s_3+s_4-s_5+s_6$

$\bar{N}=192$.

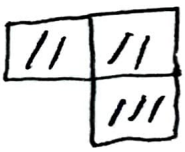
② A pair of dice, one red and the other green is rolled 6 times. We know that the ordered pairs (1,1), (1,5), (2,4), (3,6), (4,2), (4,4), (5,1) and (5,5) do not occur. What is the probability that every value on both the red die and the green die under the restrictions.





C_1: 

C_2: 

C_3: 

$r(C_1,x)=1+4x+2x^2$ $r(C_2,x)=1+x$; $r(C_3,x)=1+3x+x^2$

$r(C,x)=r(C_1,x)\times r(C_2,x)\times r(C_3,x)$

$=(1+4x+2x^2)(1+x)(1+3x+x^2)$

$r(C,x)=1+8x+22x^2+25x^3+12x^4+2x^5$

Here $n=6$, $r_0=1$, $r_1=8$, $r_2=22$, $r_3=25$, $r_4=12$, $r_5=2$ and $r_6=0$

$s_0=6!$, $s_k=(n-k)!\times r_k=(6-k)!\times r_k$

$s_1=5!\times 8$; $s_2=4!\times 22$; $s_3=3!\times 25$;

$s_4=2!\times 12$; $s_5=1!\times 2$, $s_6=0!\times 0=0$

$\bar{N}=s_0-s_1+s_2-s_3+s_4-s_5+s_6$

$=160$.

Consequently, the no. of ordered sequence of the six rolls of the dice for the Event we are interested in

$$6! \times \bar{N} = 720 \times 160$$

Since, the sample space consists of all sequence of six ordered pairs selected with the repetitions from the 28 unshaded squares of the board, the probability of this Event is

$$= \frac{720 \times 160}{(28)^6} = 0.00024.$$

Arrangements with Forbidden Positions

Suppose 'm' objects are to be arranged in 'n' places, where $n \geq m$. Suppose there are constraints under which some objects cannot occupy certain places - such places are called the forbidden positions for the said objects. The no. of ways of carrying out this task is given by the following rule:

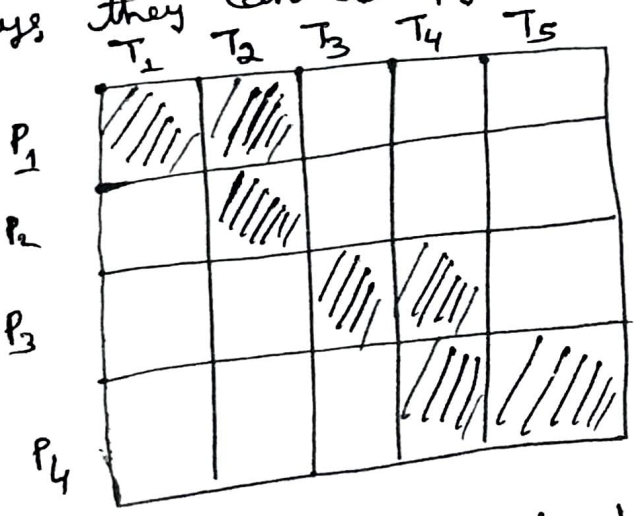
$$\bar{N} = S_0 - S_1 + S_2 - S_3 + \cdots + (-1)^n S_n$$

where $S_0 = n!$ and $S_k = (n-k)! \times r_k$, for $k = 1, 2, \cdots, n$

Here r_k is the coefficient of x^k in the rook polynomial of the board of m rows and n columns, whose squares represent the forbidden places (under the specified conditions).

① The seating arrangement of 4 persons P_i, $1 \leq i \leq 4$ at 5 different tables T_j, $1 \leq j \leq 5$ with the following restrictions as:

(i) P_1 will not sit at T_1 (or) T_2
(ii) P_2 will not sit at T_2
(iii) P_3 will not sit at T_3 (or) T_4
(iv) P_4 will not sit at T_4 (or) T_5. Find the no. of ways they can occupy the vacant chairs.

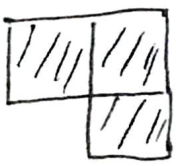


The shaded squares in the first row indicate that tables T_1 and T_2 are forbidden for P_1 and so on.

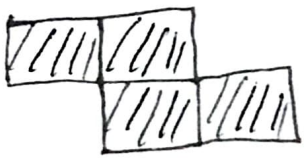
Find the rook polynomial for the shaded squares.

Here there exists two disjoint subboards:

C_1:



C_2:



At C_1: we have

$r_0 = 1 : r_1 = 3 : r_2 = 1$

$r(C_1, x) = 1 + 3x + x^2$

At C_2:

$r_0 = 1 : r_1 = 4 : r_2 = 3$

$r(C_2, x) = 1 + 4x + 3x^2$

$\therefore r(C, x) = r(C_1, x) \times r(C_2, x) = (1 + 3x + x^2) \times (1 + 4x + 3x^2)$

$\boxed{r(C, x) = 1 + 7x + 16x^2 + 13x^3 + 3x^4}$

Here $r_1 = 7 : r_2 = 16 : r_3 = 13 : r_4 = 3$.

Here $n = 5$

$S_0 = 5! = 120$. we have $S_k = (n-k)!\, r_k$

$S_1 = (5-1)! \times r_1 = 7 \times 4! = 168$

$S_2 = (5-2)! \times r_2 = 16 \times 3! = 96$

$S_3 = (5-3)! \times r_3 = 13 \times 2! = 26$

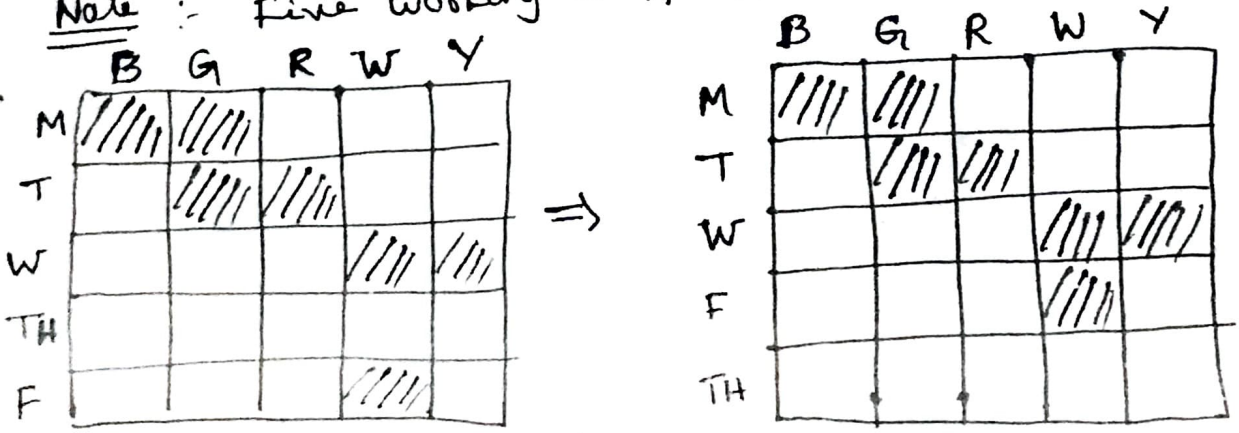
$S_4 = (5-4)! \times r_4 = 3 \times 1! = 3$

Consequently, the number of ways in which the four persons can occupy the chairs is

$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4 = 120 - 168 + 96 - 26 + 3 = 25.$

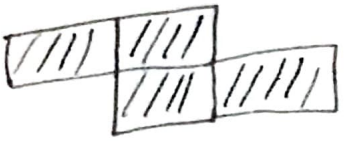
(2) An executive attending a week long business seminar has 5 suits of different colours. On Mondays he does not wear blue (or) green, on Tuesdays he does not wear red (or) green, on Wednesdays he does not wear white (or) yellow, on Fridays he does not wear white. In how many ways can he dress with the above conditions ~~sta~~ satisfied.

Note :- Five working days are taken as a week.



Find the rook polynomial for the shaded squares.

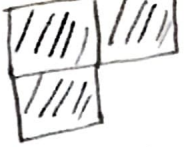
Here there exists two disjoint subboards.

C_1 : 

Here $r_0 = 1$: $r_1 = 4$

$r_2 = 3$

$r(C_1, x) = 1 + 4x + 3x^2$

C_2 : 

Here $r_0 = 1$; $r_1 = 3$; $r_2 = 1$

$r(C_2, x) = 1 + 3x + x^2$

$\therefore r(C, x) = r(C_1, x) \times r(C_2, x)$

$= [1 + 4x + 3x^2] \times [1 + 3x + x^2]$

$r(C, x) = 1 + 7x + 16x^2 + 13x^3 + 3x^4$.

Here $r_1 = 7$; $r_2 = 16$; $r_3 = 13$ and $r_4 = 3$. Also, we have $n = 5$

$S_0 = 5! = 120$, we have $S_k = (n-k)! \times r_k$

$S_1 = (5-1)! \times r_1 = 7 \times 4! = 168$

$S_2 = (5-2)! \times r_2 = 16 \times 3! = 96$

$S_3 = (5-3)! \times r_3 = 13 \times 2! = 26$

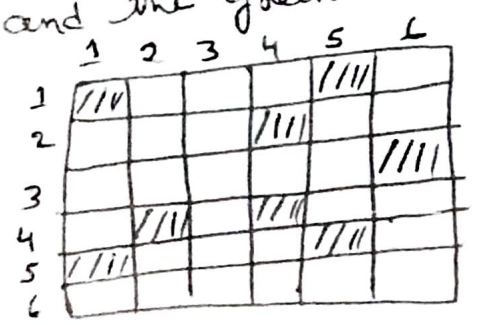
$S_4 = (5-4)! \times r_4 = 3 \times 1! = 3$; $S_5 = (5-5)! \times r_5 = 0! \times 0 = 0$

Consequently, the no. of ways he can dress with 5 color suits is

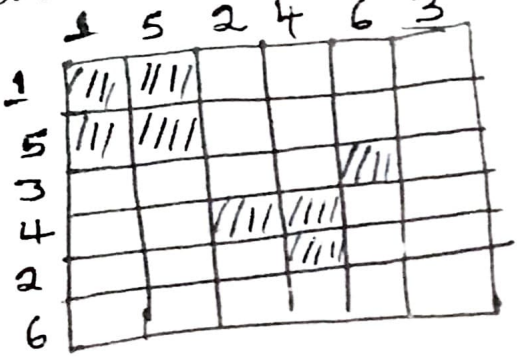
$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4 - S_5$

$= 120 - 168 + 96 - 26 + 3 - 0 = 25$.

③ A pair of dice, one red and the other green is rolled 6 times. We know that the ordered pairs (1,1), (1,5), (2,4), (3,6), (4,2), (4,4), (5,1) and (5,5) do not occur. What is the probability that every value on both the red die and the green die under the restrictions.

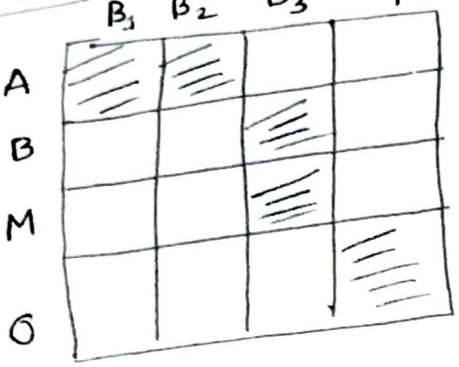


$\Rightarrow$

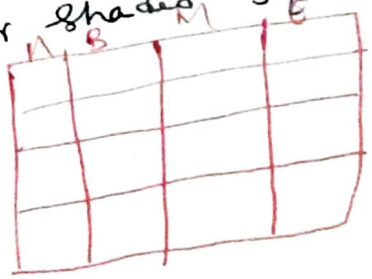


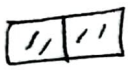
⑥ An apple, a banana, a mango and an orange are distributed to 4 boys B_1, B_2, B_3, B_4. The boy B_1 and B_2 do not wish to have apple, the boy B_3 does not want banana or mango, and B_4 refuses orange. In how many ways the distribution can be made so that no boy is displeased? displeased? no boy is displeased?

A



Find the rook polynomial for shaded squares



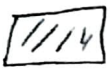
C_1: 

$r_0 = 1$: $r_1 = 2$

$r(C_1, x) = 1 + 2x$

C_2: 

$r(C_2, x) = 1 + 2x$

C_3: 

$r(C_3, x) = 1 + x$

$r(C, x) = [1+2x]^2 (1+x) = 1 + 5x + 8x^2 + 4x^3$

Here $n = 4$: $r_1 = 5$: $r_2 = 8$: $r_3 = 4$ and $r_4 = 0$

$S_0 = 4!$, Also, we have $S_k = (n-k)! \times S_k$

$S_1 = (4-1)! \times r_1 = 5 \times 3! = 30$: $S_2 = (4-2)! \times r_2 = 8 \times 2 = 16$

$S_3 = (4-3)! \times r_3 = 4 \times 1 = 4$ and $S_4 = (4-4)! \times r_4 = 0$

we have $N = S_0 - S_1 + S_2 - S_3 + S_4$

$= 24 - 30 + 16 - 4 + 0 = 6$

Five teachers T_1, T_2, T_3, T_4, T_5 are to be made class teacher for five classes, C_1, C_2, C_3, C_4, C_5, one teacher for each class. T_1 and T_2 do not wish to become the class teacher for C_1 or C_2, T_3 and T_4 don't wish to become the class teacher C_4 or C_5 and T_5 does not wish to become — " — C_3 or C_4 or C_5. In how many ways can the teacher be assigned the work without displeasing any teacher?